

Karen Kim Niet Chan Kwong Lun

Ottawa, Ontario

+1 613-700-8957

karenckl2015@gmail.com

[LinkedIn: in/karen-chan-kwong-lun](#) | [GitHub: KarenCKL](#) | [Portfolio: karenckl.github.io/my-portfolio](#)

SUMMARY OF QUALIFICATIONS

- Bilingual Software Engineering graduate (June 2026), University of Ottawa Co-op, CGPA 9.7/10, Dean's Honor Award and Merit Scholarship every term; Certified Fortinet NSE 4 – FortiOS 7.6 Administrator with direct exposure to Fortinet security technologies
- Over 16 months of co-op experience designing, developing, testing, and optimizing full-stack, frontend, and embedded software across two organizations in agile environments
- Proficient in JavaScript, TypeScript, React.js/Redux, Node.js, HTML5, CSS, C, C++, Java, Python, and C#; experienced with REST APIs, WebSocket, WebRTC, MySQL, MongoDB, and cross-platform desktop (Windows, macOS, Linux) environments
- Familiar with MERN and LAMP stacks; experienced with Docker, CI/CD, cloud tools (AWS/Azure), and real-time communication protocols including Socket.io, HTTP/2, and WebRTC
- Continuously building expertise in AI/ML (LLMs, RAG, anomaly detection), leveraging GitHub Copilot and OpenCode to accelerate development
- Fluent in English and French

EDUCATION

Bachelor of Applied Science, Software Engineering (Co-op)

2022 – 2026

University of Ottawa, Ottawa, ON

- French Immersion Stream and Engineering Management & Entrepreneurship option | CGPA: 9.7/10 | Dean's Honor Award & Merit Scholarship: All terms
- Relevant courses: Real-Time & Embedded Software Design, Secure Enterprise Network Infrastructure (Fortinet), Data Communications & Networking (Cisco), Software Engineering Capstone, Software Design & Architecture, Software Quality Assurance, Database, Operating Systems, Analysis & Design of User Interfaces

RELEVANT EXPERIENCE

Software Developer Intern

May – Aug 2025

University of Ottawa – Co-op Department, Ottawa, ON

- Engineered and implemented new frontend and backend features for the uOCompetencies web application by applying the Microsoft Blazor framework, MudBlazor components, and C#, improving system functionality and the user experience for administrative staff
- Debugged and resolved software defects across the InDEX App and notificationSender App by adapting existing C#, HTML, CSS, and VB.net code to new requirements; collaborated via GitLab using agile practices to ensure timely, high-quality delivery

Software Developer Intern

Sep – Dec 2024

Warner Bros. Discovery, Ottawa, ON

- Developed and implemented new features, resolved bugs, and wrote unit tests for the Max streaming application on Roku devices by coding in BrightScript, SceneGraph, and XML, improving application reliability and maintainability
- Reviewed pull requests from cross-functional teammates by critically analyzing code structure and logic to ensure quality standards, and tracked tasks using Jira and Confluence within a biweekly agile cycle including MergeFest, TestFest, and TeamFest

Web Developer Intern

Jan – Apr 2024

University of Ottawa – Teaching and Learning Support Service (TLSS), Ottawa, ON

- Constructed BrightSpace e-learning courses from storyboards by applying HTML, CSS, JavaScript, jQuery, and Bootstrap; collaborated with designers to produce accessible, user-friendly interfaces that met diverse learner needs

IT Support Analyst Intern & Part-Time

May 2023 – Apr 2025

University of Ottawa – Central IT Department, Ottawa, ON

- Diagnosed and resolved software and hardware issues for students and staff by configuring MFA, VPN, VMware, and cloud backups; managed device imaging via SCCM and administered virtual machines with Hyper-V to maintain reliable IT operations

Teaching Assistant / Lab Demonstrator / Corrector – ITI 1100

Jan – Apr 2026

University of Ottawa, Faculty of Engineering, Ottawa, ON

- Guided students through lab sessions and course material by clarifying complex digital systems concepts; evaluated assignments, midterms, and exams with constructive feedback; coached students requiring individualized support to strengthen their understanding

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, C#, JavaScript, TypeScript (ES6), BrightScript, Visual Basic, PHP

Frontend: HTML5, CSS3/SASS, React.js/Redux, Blazor, MudBlazor, Bootstrap, jQuery, XML, Axios, SceneGraph, Figma, PWA, WebRTC, WebSocket

Backend & Frameworks: Node.js, Express.js, Spring Boot, REST APIs, .NET, Socket.io, HTTP/2

Databases: MySQL, MongoDB, PostgreSQL, Firebase, MS SQL Server, Entity Framework

DevOps & Cloud: Docker, Git, GitHub, GitLab, Jira, Confluence, Postman, SCCM, Hyper-V, VMware, AWS/Azure (familiar)

AI & Dev Tools: GitHub Copilot, OpenCode, Claude API / Anthropic SDK, Prompt Engineering, RAG, LLM-assisted development

Testing & QA: TDD, BDD, Gherkin, unit & integration testing, Espresso

Networking & Security: Fortinet NSE 4 (FortiOS 7.6), Cisco networking, VPN, MFA, cross-platform desktop (Windows, macOS, Linux)

APPLIED PROJECTS

IoT Security Testbed (IoTa) – Software Engineering Capstone 2025 – 2026

- Designed and implemented an IoT security testbed by generating datasets across Zigbee, MQTT, Thread, and Matter protocols; built automated logging pipelines (TimescaleDB, Redis, BullMQ, Express.js) and a Grafana dashboard to visualize and analyze security events
- Explored and implemented machine learning anomaly detection on generated IoT datasets by leveraging GitHub Copilot and OpenCode to accelerate model experimentation and ML pipeline development

Full-Stack Hotel Management System Feb – Apr 2025

- Collaborated in a team of 4 to build a hotel management system using React.js, Node.js, Express.js, and PostgreSQL; connected frontend to backend using Axios and CORS, and validated all API endpoints with Postman to ensure system reliability

Full-Stack Pharmacy Prescription Management System (UYP-PMS) Sep 2025

- Developed in a team of 5 using Spring Boot (Kotlin), Angular, and PostgreSQL; implemented role-based authentication with Spring Security, Docker containerization, and validated user workflows with BDD tests written in Gherkin

Ottawa Badminton Club – Responsive Website Jun – Jul 2024

- Designed and developed a fully responsive website using React.js, Node.js, Bootstrap, HTML, CSS, SASS, JavaScript, jQuery, JSON, and XML; applied user-centered design, HCI principles, and prototyping best practices
- Achieved a 100% Google Lighthouse accessibility score by applying heuristic evaluation, WCAG-compliant design, user goal and task analysis, and visual and verbal communication strategies to deliver an inclusive, intuitive interface

Android Healthcare Appointment Scheduling & Management App Sep – Dec 2023

- Engineered an Android application in Android Studio using Java and Firebase for database management and user authentication, collaborating in a team of 6 to deliver a full appointment scheduling and management system
- Integrated multi-screen navigation via Android intents and coordinated feature merges through Git pull requests; designed UI layouts using Figma to guide front-end implementation

INTERESTS

- Open to any IT or software-related opportunity – dedicated, quick to learn, and eager to adapt, contribute, and grow within any team or technical environment
- Hardworking and detail-oriented, with a strong sense of ownership and a track record of delivering high-quality work through four co-op roles and multiple academic team projects
- Motivated by real-world impact and passionate about building reliable, accessible, and people-centered solutions – interested in real-time communication systems and cybersecurity

EXTRA-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCE

Hackathon Participant | uOttawa – Participated in hackathon events, collaborating with peers to develop innovative software solutions under time constraints.

Engineering Guide (101 Week) | Software Engineering Student Association (uOttawa SESA) Sep 2025, 2026

- Guided first-year students and completed training in first aid, sexual violence awareness, and student support.

Cook | PRD_uOttawa Club Sep 2022 – Dec 2022

- Assisted in preparing free plant-based meals from would-be food waste every week at university.

Member | High School UNESCO Club Jan 2020 – Dec 2020

- Supported an anti-drug campaign, tutored English, French, and Math, and organized activities at an elderly care home.